



Mohit Kumar Thakur

B.Tech – ECE (Minor: AI/ML)

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GitHub

LinkedIn

Portfolio

EDUCATION

University of Delhi, Faculty of Technology

B.Tech – Electronics & Communication Engineering (Minor: AI & Machine Learning)

Sep 2023 – Jul 2027

CGPA: 7.61/10 (Sem I–VI)

Mother's International Academy & Gyan Jyoti Res. Public School, Patna (CBSE)

Secondary & Senior Secondary Education

2021 & 2023

X: 91.4% | XII: 81%

TECHNICAL SKILLS AND INTERESTS

Languages: Python, C, C++

Developer Tools: Git, GitHub, Claude Code, MATLAB, Jupyter Notebook, VS Code, Linux

Frameworks/Libraries: FastAPI, React.js, PyTorch, scikit-learn, LangChain, LangGraph, Hugging Face

Cloud/Databases: PostgreSQL, Redis, Vercel

AI/ML & GenAI: RAG, Agentic AI, NLP, Neural Networks, Reinforcement Learning, GenAI, Prompt Engineering (*exploring Quantum ML*)

CS Fundamentals: OOP, Operating Systems, Data Structures & Algorithms

EXPERIENCE

Indian Institute of Information Technology (IIIT) Delhi

Research Intern – Hybrid VLC/RF Vehicular & Aerial Networks

Feb 2026 – Present

New Delhi, India

- Working under Dr. Gurinder Singh and Dr. Bohara, extended a foundational Random Forest congestion-aware framework into a three-stage intent-aware pipeline (MLP intent classification, per-intent reliability gating, AHP-weighted utility ranking), benchmarked against AODV/DSDV baselines across three congestion regimes, and is revising the manuscript per reviewer feedback with an added ablation study for conference submission.

RESEARCH WORK

Hierarchical Intent-Aware Routing for Hybrid VLC/RF VANET-FANET Networks

Python, scikit-learn, MATLAB, AHP

2026

[GitHub](#)

- Designed a cascaded MLP intent-classification + per-intent reliability-gating + AHP-weighted utility-ranking pipeline (Saaty pairwise matrices, consistency-ratio verified); benchmarked against AODV/DSDV baselines across three congestion regimes and open-sourced the full reproducibility package.

ML-Driven Utility-Based Hybrid VLC-RF Routing for Congestion-Aware UAV-Assisted VANETs

Python, MATLAB, scikit-learn

2026

[GitHub](#)

- Trained a 350-tree Random Forest classifier on 7 cross-layer features over 20,000 samples with a 6-criteria multi-objective utility function, achieving up to 62.3% congestion-index reduction and 59.5% E2E delay reduction over the AODV baseline.

PROJECTS

AI Research Agent

FastAPI, Celery, Redis, PostgreSQL, Groq LLM, React, JWT

2026

[Live Demo](#)

[GitHub](#)

- Built a full-stack research assistant: FastAPI hands queries to a Celery/Redis worker that web-searches and summarizes with a Groq LLM, returning sourced reports to a React 18 dashboard behind JWT auth; also prototyped standalone RAG/agent modules (ChromaDB, ReAct loop).

End-to-End Weather Data Pipeline with ML Prediction

Python, PostgreSQL, SQLAlchemy, scikit-learn, XGBoost, FastAPI

2026

[Live Demo](#)

[GitHub](#)

- Built a pipeline ingesting Open-Meteo data into PostgreSQL (8,900+ records), engineered 18 features, and compared Linear Regression, Random Forest, and XGBoost – best model achieved $R^2 = 0.9975$, served via a FastAPI REST API deployed live on Render.

DevNest – Campus Communication & Opportunity Platform

React, TypeScript, Node.js, Express, MongoDB, Socket.IO

2026

[GitHub](#)

- Built a full-stack platform (React/TypeScript, Node.js/Express + MongoDB) unifying auth, an opportunity dashboard, a learning module, and a moderated community system, with JWT/Google OAuth, real-time Socket.IO updates, and a CRON-scheduled scraper aggregating Unstop/Devpost listings.

MediGenie – AI Health Companion for Disease Prediction

React, FastAPI, Express.js, MongoDB, scikit-learn

2025

[GitHub](#)

- Built a JWT-secured health companion (React, FastAPI/Express, MongoDB) predicting disease risk across 8 disease datasets with a Random Forest classifier, using SMOTE and GridSearchCV to improve accuracy by approximately 15% over baseline.

ACHIEVEMENTS & CERTIFICATIONS

Hackathon Winner – Astraaya 1.0: Built ResQRoute, emergency-vehicle coordination platform; 1st place

2026

Competitive Programming: GeeksforGeeks Institute Rank 4th (FoT, DU) | LeetCode Contest Rating 1499

Ongoing

NPTEL – Cloud Computing

Apr 2026

Deloitte Forage Virtual Job Simulations – Data Analytics, Technology, Cyber Security

Jun 2025

TCS iON Certifications – Master Data Management & Soft Skills (100%)

Jun 2025